QISKIT FALL FEST 2026 · VNQFF-02

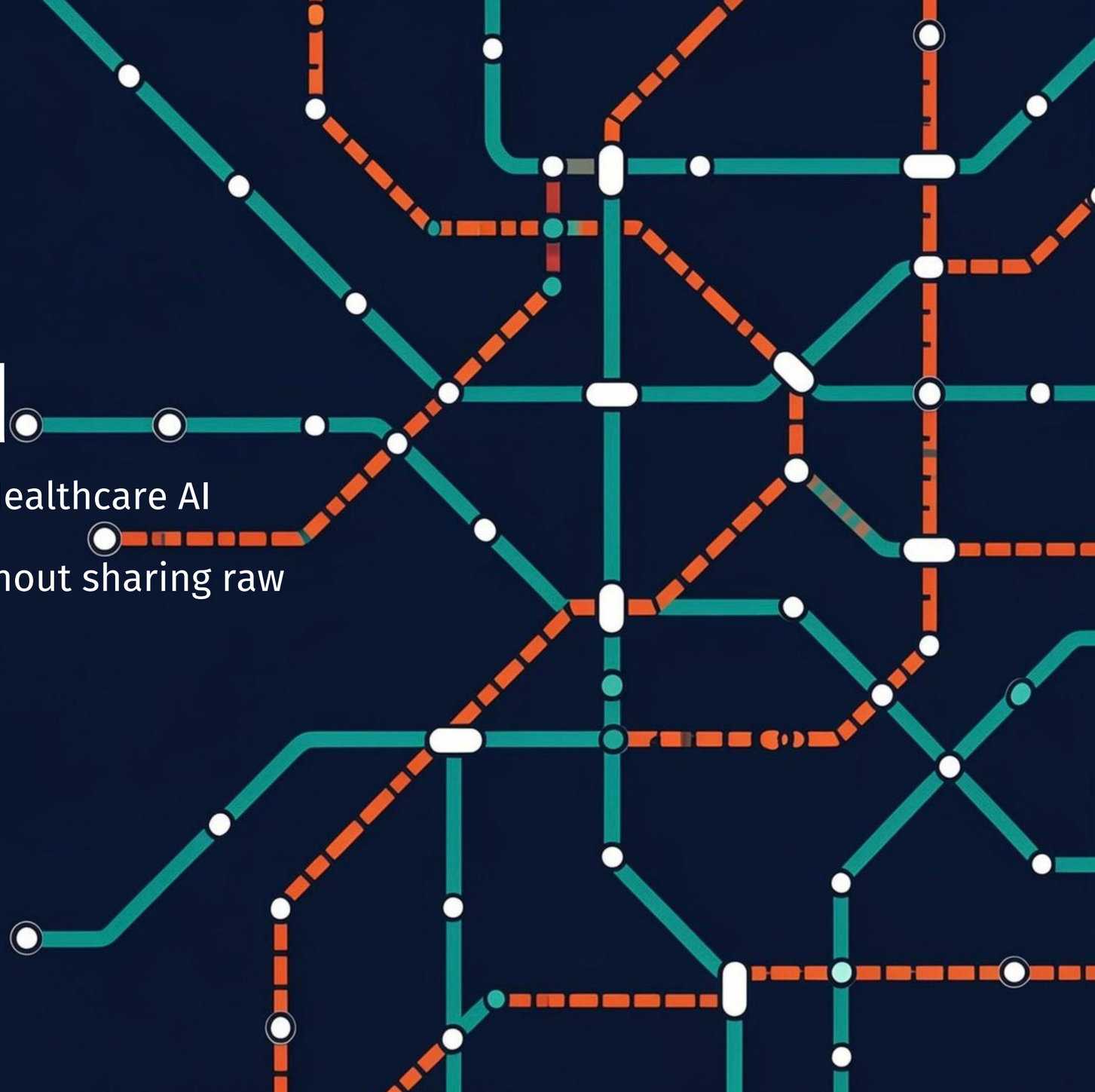
Q-HealthGuard

Privacy-Preserving Federated Quantum Healthcare AI

Collaborative healthcare intelligence without sharing raw patient data.

QISKIT

EPIC ENGINEERS



EXECUTION FLOW

Train locally. Share parameters. Improve globally.

Multiple hospitals collaboratively train healthcare prediction models without directly sharing raw patient records.

TRIAD SYNTHESIS

Federated Averaging + Qiskit QML Kernels + Differential Privacy.

01 **Hospital A, B, C**

Patient data remains within local hospital premises.

02 **Local quantum training**

Each client trains a Qiskit circuit model locally.

03 **Privacy layer**

Clip updates and add Differential Privacy noise.

04 **Parameter-only exchange**

Only encrypted quantum circuit weights are transmitted.

05 **Global model**

Federated Aggregation with Federated Averaging/FedAvg.

IMPLEMENTATION STACK

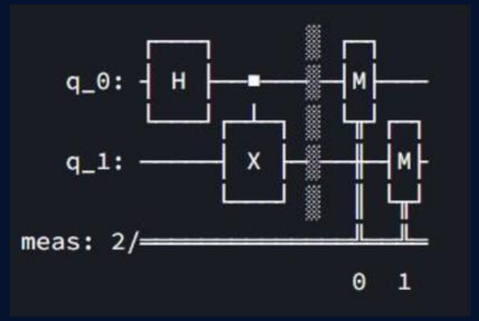
Built from open tools and a small circuit

TECHNOLOGY STACK

Frontend — Streamlit
Core language — Python 3.10+
Data & ML — Pandas · Scikit-learn
Quantum SDK — Qiskit 1.x · Qiskit QML
Simulation — Qiskit Aer
Cloud quantum — IBM Quantum Compute

METHODOLOGY PIPELINE

1. **Patient preprocessing** — Feature normalization into [0, π] angles.
2. **Qiskit QML encoding** — State mapping with an entangling ansatz.
3. **Local client training** — Quantum kernel classification optimization.
4. **Privacy + FedAvg** — Gradient clipping and secure parameter fusion.



Qiskit Aer supports fast local exact and noisy runs.

FEASIBILITY AND VIABILITY

Four risks, each with a bounded mitigation

The prototype stays feasible by keeping the circuit shallow, the data local, and the exchanged updates protected.

POTENTIAL CHALLENGE & RISK	STRATEGY TO OVERCOME · STATUS
Small quantum hardware & noise	Qiskit Aer local noise simulation + shallow depth circuits · **Mitigated**
Privacy leakage in model updates	Update clipping threshold + Differential Privacy noise addition · **Protected**
Heterogeneous hospital datasets	Standardized local feature scaling into unified quantum angle ranges · **Standardized**
Computational bottlenecks	Lightweight 4-qubit parameterized classification ansatz · **Optimized**

TECHNICAL FEASIBILITY

Qiskit Aer provides local noise and state vector simulation without requiring cloud QPU queues during development.

OPERATIONAL FEASIBILITY

Hospitals retain full data custody. Prototype validates on 3 simulated local clients using synthetic clinical data.

ECONOMIC VIABILITY

Built with 100% open-source software stack. IBM Quantum Compute cloud hardware used strictly for verification runs.

IMPACT AND BENEFITS

Privacy boundary, shared value across four audiences

Q-HealthGuard keeps patient data local while making cross-institutional intelligence possible.

POTENTIAL AUDIENCE IMPACT

Hospitals & Networks

Collaborative cross-institutional intelligence without GDPR/HIPAA violation liability.

Healthcare Professionals

More generalizable clinical risk prediction models supported by diverse demographic cohorts.

Research Institutions

Benchmark real-world quantum kernel advantages in privacy-preserving medical diagnostics.

BENEFITS OF THE SOLUTION

Absolute privacy

Guarantees patient confidentiality across healthcare boundaries.

Zero central infra

Eliminates expensive redundant central databases and pipelines.

QML advantage

Validates quantum kernel mapping in real-world clinical AI.

Lightweight compute

Lowers energy footprint compared to massive centralized data transfers.

RESEARCH AND REFERENCES

Grounded in IBM Quantum tools and established methods

PRIMARY IBM QUANTUM LITERATURE · USE IN Q-HEALTHGUARD

Qiskit Documentation (Qiskit SDK 1.x) — Quantum circuits, parameterized rotation gates, and transpilation

IBM Quantum Kernel Training (QKT) — Feature mapping to high-dimensional Hilbert space for classification

Qiskit Aer Simulation Guides — Exact and realistic device noise modeling with custom noise matrices

IBM Quantum Compute Client (Runtime API) — Cloud primitives for hardware execution and verification

KEY THEORETICAL FOUNDATIONS

Federated Learning (FedAvg)

McMahan et al. · Multi-institutional model aggregation

Quantum-enhanced feature spaces

Havlíček et al. · Supervised learning with quantum kernels

Differential Privacy (DP)

Dwork et al. · Rigorous privacy bounds on parameter sharing

RESOURCE PORTAL

quantum.cloud.ibm.com/docs

Qiskit Fall Fest 2026 Submission · Jovial Korivi